



(<http://www.pieriandata.com>)

Groupby

The groupby method allows you to group rows of data together and call aggregate functions

```
In [31]: import pandas as pd
# Create dataframe
data = {'Company': ['GOOG', 'GOOG', 'MSFT', 'MSFT', 'FB', 'FB'],
        'Person': ['Sam', 'Charlie', 'Amy', 'Vanessa', 'Carl', 'Sarah'],
        'Sales': [200, 120, 340, 124, 243, 350]}
```

```
In [32]: df = pd.DataFrame(data)
```

```
In [33]: df
```

```
Out[33]:
```

	Company	Person	Sales
0	GOOG	Sam	200
1	GOOG	Charlie	120
2	MSFT	Amy	340
3	MSFT	Vanessa	124
4	FB	Carl	243
5	FB	Sarah	350

** Now you can use the .groupby() method to group rows together based off of a column name. For instance let's group based off of Company. This will create a DataFrameGroupBy object:**

```
In [34]: df.groupby('Company')
```

```
Out[34]: <pandas.core.groupby.DataFrameGroupBy object at 0x113014128>
```

You can save this object as a new variable:

```
In [35]: by_comp = df.groupby("Company")
```

And then call aggregate methods off the object:

```
In [36]: by_comp.mean()
```

```
Out[36]:
```

	Sales
Company	
FB	296.5
GOOG	160.0
MSFT	232.0

```
In [37]: df.groupby('Company').mean()
```

```
Out[37]:
```

	Sales
Company	
FB	296.5
GOOG	160.0
MSFT	232.0

More examples of aggregate methods:

```
In [38]: by_comp.std()
```

```
Out[38]:
```

	Sales
Company	
FB	75.660426
GOOG	56.568542
MSFT	152.735065

```
In [39]: by_comp.min()
```

```
Out[39]:
```

	Person	Sales
Company		
FB	Carl	243
GOOG	Charlie	120
MSFT	Amy	124

```
In [40]: by_comp.max()
```

```
Out[40]:
```

	Person	Sales
Company		
FB	Sarah	350
GOOG	Sam	200
MSFT	Vanessa	340

```
In [41]: by_comp.count()
```

```
Out[41]:
```

	Person	Sales
Company		
FB	2	2
GOOG	2	2
MSFT	2	2

In [42]: `by_comp.describe()`

Out[42]:

Sales											
Company											
FB	count	2.000000									
	mean	296.500000									
	std	75.660426									
	min	243.000000									
	25%	269.750000									
	50%	296.500000									
	75%	323.250000									
	max	350.000000									
GOOG	count	2.000000									
	mean	160.000000									
	std	56.568542									
	min	120.000000									
	25%	140.000000									
	50%	160.000000									
	75%	180.000000									
	max	200.000000									
MSFT	count	2.000000									
	mean	232.000000									
	std	152.735065									
	min	124.000000									
	25%	178.000000									
	50%	232.000000									
	75%	286.000000									
	max	340.000000									

In [43]: `by_comp.describe().transpose()`

Out[43]:

Company		FB										GOOG		MSFT	
		count	mean	std	min	25%	50%	75%	max	count	mean	...	75%	count	mean
Sales		2.0	296.5	75.660426	243.0	269.75	296.5	323.25	350.0	2.0	160.0	...	180.0	2.0	232.0

1 rows × 24 columns



```
In [44]: by_comp.describe().transpose()['GOOG']
```

```
Out[44]:
```

	count	mean	std	min	25%	50%	75%	max
Sales	2.0	160.0	56.568542	120.0	140.0	160.0	180.0	200.0

Great Job!